

Nonlinear Models in Financial Econometrics

TAR, Markov Switching, Nonlinearity Tests, Bootstrap and Forecast Evaluation

FI362 — Advanced Financial Econometrics

Applications to the S&P 500

July 10, 2026

Roadmap

- ① Why do we need nonlinear models?
- ② Threshold Autoregressive (TAR) models
- ③ Markov switching models
- ④ Testing for nonlinearity
- ⑤ Parametric bootstrap
- ⑥ Forecasting
- ⑦ Forecast evaluation
- ⑧ Putting it together on the S&P 500

How to read these slides

Every slide does one job. Most come in pairs.

The idea

One sentence of plain English. If you remember nothing else, remember this.

The mathematics

The same statement, written precisely.

The trap

The mistake that the mathematics exists to prevent.

The market has moods

The idea

In a calm market, a 1% move tomorrow is a surprise. In a panic, it is Tuesday.

An ARMA model for the S&P 500 return r_t says

$$r_t = \mu + \sum_{i=1}^p \phi_i r_{t-i} + a_t, \quad a_t \sim \text{WN}(0, \sigma^2).$$

Look at what is *not* allowed to move: μ , every ϕ_i , and σ^2 .

They are the same number in 2007 and in October 2008.

The model has exactly one personality.

Three things a linear model cannot produce

① Regime-dependent dynamics.

Persistence during a crash must equal persistence in a calm market.

② Asymmetric cycles.

Falls and recoveries are forced to have the same shape.

③ Limit cycles and multiple equilibria.

A stationary ARMA always drifts back to one point, μ , forever.

Each of the next two model families buys back exactly one of these.

Nonlinear where? Two different questions

The idea

“Where is the return heading?” and “how uncertain is it?” are separate questions. A model can be nonlinear in either one.

$$r_t = \underbrace{g(\mathcal{F}_{t-1})}_{\text{where it is heading}} + \underbrace{h(\mathcal{F}_{t-1})}_{\text{how uncertain}} \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} (0, 1),$$

where $\mathcal{F}_{t-1} = \{r_{t-1}, r_{t-2}, \dots\}$ is everything you knew yesterday.

Which model does which

	g (the mean)	h (the volatility)
ARMA	linear	constant
GARCH	linear	nonlinear
TAR	nonlinear	constant or switching
Markov switching	nonlinear	switching

The trap

GARCH *is* a nonlinear model — but it is linear in the **mean**.

Tests for ARCH tell you about h and nothing whatsoever about g . This one distinction decides almost every empirical question in this lecture.

Two ways to let the mood matter

The idea

Either you can *see* which mood the market is in, or you can only *guess*. That single difference produces the two model families.

TAR: the mood is visible

A rule you can check yourself:

$$s_t = \mathbf{1}\{r_{t-d} > c\}.$$

Yesterday's return was above c , or it was not. No ambiguity, ever.

A thermostat.

Markov: the mood is hidden

An unobserved coin, weighted by yesterday:

$$\Pr(s_t = j \mid s_{t-1} = i) = p_{ij}.$$

You never see s_t . You compute $\Pr(s_t = j \mid \mathcal{F}_t)$.

Diagnosing a fever from symptoms.

The SETAR model: two AR models and a switch

The idea

Run one AR model on quiet days and a different AR model on wild days. Which day it is gets decided by a number you already observed.

$$r_t = \begin{cases} \phi_0^{(1)} + \sum_{i=1}^{p_1} \phi_i^{(1)} r_{t-i} + \sigma_1 \varepsilon_t, & r_{t-d} \leq c, \\ \phi_0^{(2)} + \sum_{i=1}^{p_2} \phi_i^{(2)} r_{t-i} + \sigma_2 \varepsilon_t, & r_{t-d} > c. \end{cases}$$

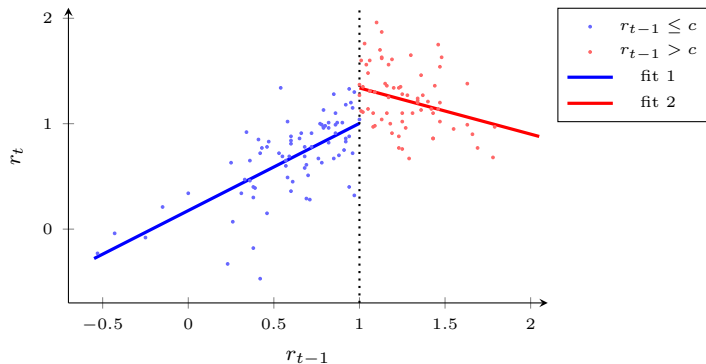
Reading the SETAR equation

- c is the **threshold** — the dividing line.
- d is the **delay** — how far back you look to decide.
- “Self-exciting” means the series decides its own regime. Nothing external is needed.
- Two straight lines glued together. Each piece is linear; the join makes the whole thing nonlinear.
- A different σ_j in each regime gives volatility asymmetry — *with no GARCH term anywhere*.

On the S&P 500

$d = 1$ and $c \approx 0$ reads as: “after a down day” versus “after an up day”. Typically the down-day regime has bigger σ and stronger snap-back — the leverage effect, showing up in the mean equation.

What the threshold does to the data



Below $c = 1$ the slope is $+0.83$: **momentum**, the series climbs. Above c the slope is -0.44 : **reversal**, the series is pulled back.

Why one line will not do

The idea

A single regression line through that cloud would report an average slope of roughly 0.3 — and would describe neither half of the data.

- Momentum (+0.83) and reversal (−0.44) are not close to each other.
- Their average is a number that occurs in neither regime.
- The linear model is not *wrong on average*. It is wrong everywhere, in opposite directions.

*That scatter plot is the entire SETAR model.
Everything that follows is how to find the dotted line.*

Switch off the noise and watch the machine run

The idea

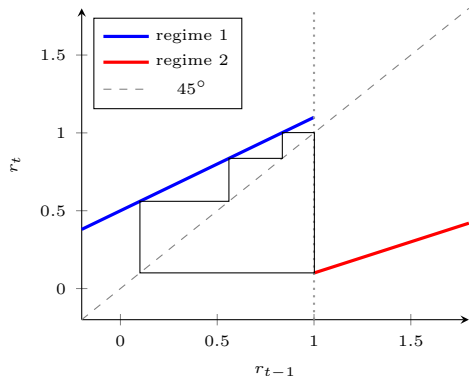
Delete the shocks. What is left is a machine mapping today onto tomorrow. Where does it come to rest?

Set $\varepsilon_t = 0$, so that $r_t = f(r_{t-1})$ with

$$f(y) = \begin{cases} 0.5 + 0.6y, & y \leq 1, \\ -0.3 + 0.4y, & y > 1. \end{cases}$$

A resting point needs $f(y) = y$: the line must touch the 45° line *inside its own regime*.

Both resting points are imaginary



Regime 1 wants to rest at

$$\frac{0.5}{1 - 0.6} = 1.25.$$

But $1.25 > 1$, where regime 1 does not apply.

Regime 2 wants to rest at

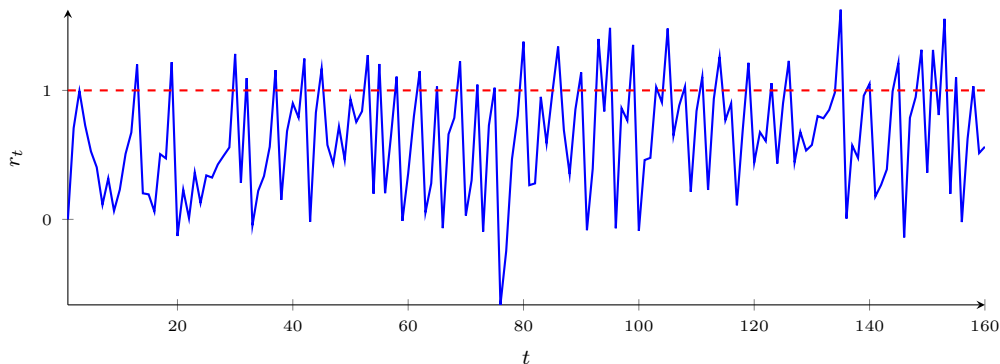
$$\frac{-0.3}{1 - 0.4} = -0.5.$$

But $-0.5 \leq 1$, where regime 2 does not apply.

Both are **virtual** fixed points. The machine is chased around a **period-4 cycle**, forever.

No stationary ARMA can do this.

The same machine, with the noise switched back on



Simulated SETAR(2; 1, 1), threshold $c = 1$ (dashed).

Look at the shape: the rises are long and gentle, the falls are short and violent.

That asymmetry is the point. An ARMA gives you symmetric wiggles.

Stationarity: a bubble regime is allowed

The idea

One regime may be explosive, provided the other regime drags the series back. Stability is a property of the *pair*.

Sufficient condition (two regimes, $p = 1$)

$$\phi_1^{(1)} < 1, \quad \phi_1^{(2)} < 1, \quad \phi_1^{(1)} \phi_1^{(2)} < 1.$$

Note what is *missing*: nobody said $|\phi_1^{(j)}| < 1$ in each regime.

Why economists like that condition

- $\phi_1^{(1)} = 1.2$ — a bubble that feeds itself — is perfectly allowed.
- It only needs $\phi_1^{(2)}$ small enough that $1.2 \times \phi_1^{(2)} < 1$: the correction regime must be strong enough to clean up after it.
- So SETAR formalises “bubble, then correction” **inside a stationary model**. You do not have to abandon stationarity to talk about bubbles.

Caveat

For general p there is no clean necessary-and-sufficient condition. Simulate the skeleton and look at it.

Estimating the threshold: try every candidate

The idea

If you knew c , this would be two ordinary regressions. You don't, so try every c and keep the one that fits best.

$$\hat{c} = \arg \min_{c \in \mathcal{C}} \hat{\sigma}^2(c) = \arg \min_{c \in \mathcal{C}} \frac{1}{T} \sum_t \hat{a}_t(c)^2.$$

For each candidate c : split the sample into $\{r_{t-d} \leq c\}$ and $\{r_{t-d} > c\}$, run OLS on each half, add up the squared residuals. Pick the winner.

That is the entire estimator.

Two practicalities, one surprise

Trimming

$\mathcal{C}$ runs only over the middle $[\pi, 1 - \pi]$ quantiles of r_{t-d} , with $\pi \approx 0.15$, so that neither regime is left with three observations.

Arranged autoregression (Tsay)

Sort the rows by r_{t-d} first. Each regime then becomes a contiguous block, and recursive least squares makes the whole grid search cheap.

The surprise

$\hat{c}$ converges at rate T . The slopes converge only at rate $\sqrt{T}$.

The threshold is estimated *so much more precisely* than everything else that you may report slope standard errors as if c were known. **Never the reverse.**

Choosing d , p_j , and how many regimes

The idea

Search over the small integers by AIC.

$$\text{AIC}(p_1, p_2, d) = \sum_{j=1}^2 \left[n_j \log \hat{\sigma}_j^2 + 2(p_j + 1) \right],$$

minimised jointly over (p_1, p_2, d) , with $\hat{c}$ profiled out inside the loop.

This is fine for choosing p and d .

It is not fine for deciding whether the series is nonlinear at all.

Why AIC cannot test linearity

The trap

You chose c by picking the best of two hundred candidates. Of course the SETAR fits better — it *searched*.

The analogy

The tallest of 200 randomly chosen people is taller than average.

That tells you nothing about whether the population is unusually tall.

To learn anything, you must first know how tall the tallest-of-200 *usually* is.

That reference distribution is the bootstrap. It arrives in two sections' time.

Hamilton's model: the mood you cannot see

The idea

Assume there are two worlds, each with its own mean and its own volatility. A hidden coin decides which world you wake up in — and the coin is biased towards staying put.

$$P = \begin{pmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{pmatrix}, \quad p_{ij} = \Pr(s_t = j \mid s_{t-1} = i),$$

$$r_t = \mu_{s_t} + \sum_{i=1}^p \phi_i^{(s_t)} r_{t-i} + \sigma_{s_t} \varepsilon_t, \quad \varepsilon_t \sim N(0, 1).$$

Row i of P says: “if I am in world i today, here is where I go tomorrow.” Rows sum to one.

Making P human-readable

Two numbers turn a matrix of probabilities into English.

How long do moods last?

$$\mathbb{E}[\text{duration of regime } i] = \frac{1}{1 - p_{ii}}.$$

What share of days does each mood occupy?

$$\pi_1 = \frac{1 - p_{22}}{2 - p_{11} - p_{22}}, \quad \pi_2 = 1 - \pi_1.$$

With $p_{11} = 0.95$ and $p_{22} = 0.80$: moods last 20 and 5 days, and the market spends 80% of its life calm.

The Hamilton filter is guessing, then correcting

The idea

Before you see today's return, you have a hunch about today's mood. After you see it, you update the hunch. Once per day, forever.

Let $\xi_{t|t}$ collect $\Pr(s_t = j \mid \mathcal{F}_t)$, and η_t collect $f(r_t \mid s_t = j, \mathcal{F}_{t-1})$ — “how plausible is today's return, if I were in world j ?”

Predict, then apply Bayes

$$\underbrace{\xi_{t|t-1} = P' \xi_{t-1|t-1}}_{\text{yesterday's belief, pushed forward}}, \quad \xi_{t|t} = \underbrace{\frac{\xi_{t|t-1} \odot \eta_t}{\mathbf{1}'(\xi_{t|t-1} \odot \eta_t)}}_{\text{prior} \times \text{likelihood, renormalised}}.$$

The likelihood falls out for free

The denominator of the Bayes step is the probability of today's return *before* you knew the mood. So:

$$\ell(\theta) = \sum_{t=1}^T \log[\mathbf{1}'(\xi_{t|t-1} \odot \eta_t)].$$

- Run the filter once. You get the regime probabilities *and* the log-likelihood.
- Maximise ℓ directly (reparametrising σ_j by log and p_{ij} by logit), or use EM.
- The E-step of EM is the Kim (1994) backward pass:

$$\xi_{t|T} = \xi_{t|t} \odot [P(\xi_{t+1|T} \oslash \xi_{t+1|t})].$$

A -3% day: what the two worlds say

Take $\mu = (0.05, -0.10)$, $\sigma = (0.80, 2.50)$, prior $\xi_{t|t-1} = (0.80, 0.20)$. You observe $r_t = -3.00\%$.

$$\eta_{t,1} = \frac{1}{0.80} \phi\left(\frac{-3.00-0.05}{0.80}\right) = 3.48 \times 10^{-4}, \quad \eta_{t,2} = \frac{1}{2.50} \phi\left(\frac{-3.00+0.10}{2.50}\right) = 8.14 \times 10^{-2}.$$

A -3% day is 3.81 standard deviations from calm, but only 1.16 from turbulent.

$$\textit{Evidence ratio: } \eta_{t,2}/\eta_{t,1} = 234.$$

One bad day flips the model's mind

$$\xi_{t|t} \propto \begin{pmatrix} 0.80 \\ 0.20 \end{pmatrix} \odot \begin{pmatrix} 3.48 \times 10^{-4} \\ 8.14 \times 10^{-2} \end{pmatrix} \implies \xi_{t|t} = \begin{pmatrix} 0.0168 \\ 0.9832 \end{pmatrix}.$$

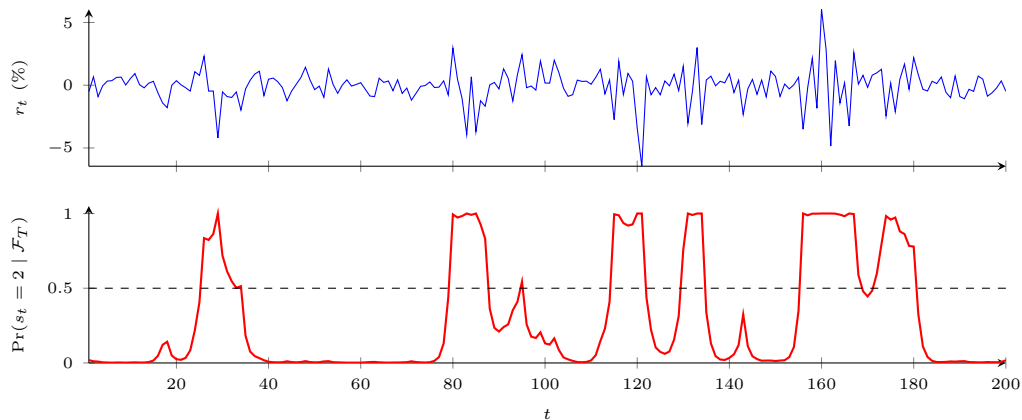
Read it as odds

Prior odds $0.20/0.80 = 0.25$. Times evidence 234. Posterior odds 58.5, i.e. 98.3%.

But wait for the smoother

If the next five days are calm, the *smoothed* probability falls back. A single outlier followed by calm is better explained as a fat tail than as a crisis that ended immediately. The filter knows only the past; the smoother sees the whole sample.

The model dates its own crises



Simulated data. Nobody told the model where the crises were — it read them off the returns.

On real S&P 500 data this red line should light up in 2008, March 2020 and 2022. If it does not, something is wrong.

TAR vs. Markov switching, side by side

	TAR / SETAR	Markov switching
Who decides the regime?	the data, visibly	a hidden coin
Switch	sharp, deterministic	random
You learn the regime	exactly	as a probability $\xi_{t t}$
Estimation	OLS + grid search	MLE / EM (Hamilton filter)
Forecast $h > 1$	simulate paths	$\xi'_{T T} P^h$, then mix
Good story for	bubbles, arbitrage bands	bull/bear, crisis dating

They share one disease

Under H_0 of linearity...

- in a TAR, the threshold c means **nothing**;
- in a Markov switching model, p_{11} and p_{22} mean **nothing**.

The likelihood does not depend on them at all.

A parameter that vanishes under the null wrecks the usual χ^2 theory.

That is the next section, and it is the most important one in the lecture.

First ask: which nonlinearity are you hunting?

The idea

“Is this series nonlinear?” is too vague to answer. Nonlinear *where* — in the mean, or in the variance?

Test	What it assumes is true	What it can smell
McLeod–Li (Q on $\hat{a}_t^2$)	no dependence in a_t^2	ARCH; variance nonlinearity
Engle ARCH-LM	no ARCH	conditional heteroskedasticity
Keenan / Tsay F	the mean is linear	curvature in the mean
BDS	residuals are i.i.d.	<i>anything at all</i>
Tsay (1989) threshold F	the model is $AR(p)$	a threshold in the mean
Hansen sup-LR	the model is $AR(p)$	a threshold, c unknown

Order matters

- ① Fit an ARMA.
- ② Test for ARCH.
- ③ **If ARCH is there, filter it out with GARCH** — *before* you interpret any BDS or threshold rejection as evidence about the mean.

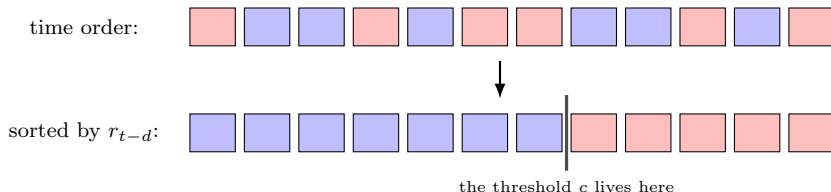
The trap

Skip step 3 and a volatility effect will walk into your paper wearing a threshold costume. BDS, Tsay's F and sup-LR all reject against a *general* alternative. They cannot tell you which alternative.

Tsay's trick: stop reading the data in time order

The idea

Time order tells you when things happened. Sorted order tells you which regime they were in.



Sorting does not change the data. It changes where the regimes sit: they stop being interleaved and become the front and the back of a queue.

Walk down the queue

Recursive least squares

Fit on the first m observations. Predict the $(m+1)$ -th. Record the error. Absorb it. Move on.

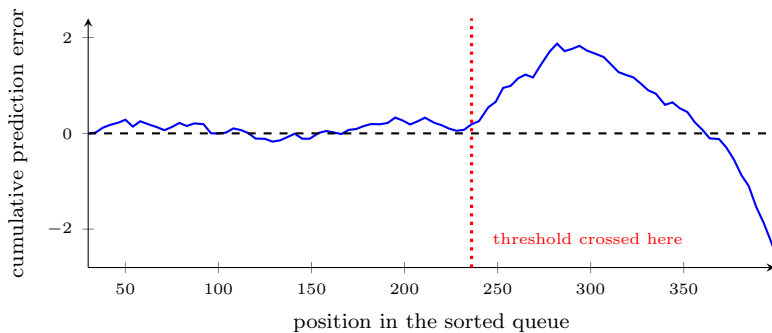
Why this earns you an ordinary F table

While you are still inside regime 1, the model you keep updating is the *right* model, so the prediction errors are pure noise.

The moment you cross c , you are predicting regime-2 observations with a regime-1 model, and the errors turn systematic.

The test never has to know where c is. It only has to notice that something broke.

The break, made visible



Up to observation 236 the cumulative error wanders around zero: one regime, one model, pure noise.

Past 236 the regime-1 model starts predicting regime-2 observations. The errors stop cancelling.

Tsay's F statistic

The idea

A formal way of asking: *is that excursion bigger than luck?*

Regress the recursive prediction errors $\hat{e}_t$ on $(1, r_{t-1}, \dots, r_{t-p})$ and test that all coefficients are zero:

$$\hat{F}(p, d) = \frac{(\text{SSR}_0 - \text{SSR}_1)/(p+1)}{\text{SSR}_1/(T_{\text{eff}} - p - 1)} \stackrel{a}{\sim} F(p+1, T_{\text{eff}} - p - 1) \quad \text{under } H_0.$$

The price

The threshold c never appears anywhere, which is why an ordinary F distribution applies. You pay for that convenience in **power**, when the true c sits far from the median.

The Davies problem

The idea

If linearity is true, the threshold c is a parameter for a thing that does not exist.

$$H_0: \text{linear AR}(p) \\ \phi^{(1)} = \phi^{(2)}$$

Under H_0 , c is **irrelevant**: the likelihood is completely flat in c .

$$H_1: \text{SETAR} \\ \text{with threshold } c$$

Under H_1 it matters. So c exists **only under the alternative**.

You cannot estimate c under the null, and the χ^2 table does not know that you went looking for it.

Why the χ^2 table is the wrong ruler

We are forced to test with

$$\sup_{c \in \mathcal{C}} \text{LR}(c)$$

— the best threshold we could find anywhere on the grid.

Consequence

The best-of-many is bigger than a typical one.

So this statistic is **not** χ^2_{p+1} . Its distribution sits to the *right*, and it depends on the data-generating process and on the grid $\mathcal{C}$.

Use χ^2 critical values and you will reject linearity far too often.

Markov switching has the identical disease: under H_0 , p_{11} and p_{22} are meaningless.

If you cannot look up the null, simulate it

The idea

Build many fake datasets in which linearity is *true by construction*. Hunt for a threshold in each one, exactly as you did with the real data.

Now you know how big $\sup_c \text{LR}$ gets **by luck alone**.

This is the “how tall is the tallest of 200 people, usually?” question, answered by measuring 999 groups of 200.

The bootstrap algorithm

Testing $\text{AR}(p)$ against SETAR with unknown c

- ① Fit the **restricted** model, $\text{AR}(p)$, giving $\hat{\theta}_0 = (\hat{\mu}, \hat{\phi}, \hat{\sigma}^2)$.
- ② Fit the SETAR by grid search; record $\widehat{\text{LR}} = \sup_c \text{LR}(c)$.
- ③ For $b = 1, \dots, B$: simulate $\{r_t^{*(b)}\}$ from the $\text{AR}(p)$ at $\hat{\theta}_0$, then **re-run the entire procedure** on it — including the grid search — to get $\text{LR}^{*(b)}$.
- ④ Compute

$$p^* = \frac{1 + \#\{b : \text{LR}^{*(b)} \geq \widehat{\text{LR}}\}}{1 + B}.$$

Use $B = 999$ or 1999 , so that $\alpha(1 + B)$ is a whole number.

The one mistake that ruins everything

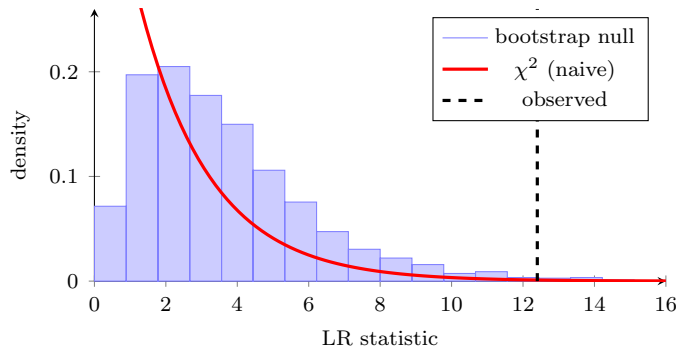
The trap

If you fix $c = \hat{c}$ inside the loop to save time, you never *searched* in the fake data.

- What you simulated is $\text{LR}(\hat{c})$, a pointwise statistic — which *is* χ^2 .
- Since $\sup_c \text{LR}^*(c) \geq \text{LR}^*(\hat{c})$ always, the simulated null comes out too small.
- Its upper quantiles are too low, so p^* is too small.
- You will find thresholds in data *you generated yourself as linear*.

The search is part of the statistic. The bootstrap must replicate the search.

The two distributions, drawn



Searching over c drags the null distribution to the right, so its 95% point sits far above the χ^2 one. Here $\widehat{\text{LR}} = 12.4$ and 17 of $B = 1999$ fakes beat it, so $p^* = 18/2000 = 0.009$. The naive χ^2_2 cutoff is only 5.99.

Which fake data should you build?

The idea

Your p -value is only as honest as the fake world you compared it against.

Three recipes for the shocks

- **Parametric:** $\varepsilon^* \sim N(0, 1)$.
Efficient *if* returns are normal. They are not.
- **Residual:** resample $\hat{a}_t$ i.i.d.
Keeps the fat tails. Assumes independence.
- **Wild:** $a_t^* = \hat{a}_t \eta_t$, with η_t i.i.d. mean 0, variance 1.
Keeps the volatility clusters.

Why this matters on the S&P 500

The trap

Daily returns cluster: quiet weeks, then violent weeks. Shuffling residuals i.i.d. **destroys** that.

The chain of consequences:

- ① The fake data are too placid.
- ② So extreme values of r_{t-d} are too rare.
- ③ So large values of $\sup_c \text{LR}$ are too rare.
- ④ So the simulated null is too tight, and the test **over-rejects**.

You will announce a threshold. It was GARCH.

Fix: bootstrap under an AR–GARCH null, or use the wild bootstrap, or Hansen’s fixed-regressor bootstrap. And record the seed and B .

Tomorrow is easy

The idea

Tomorrow's regime is already decided by data you have in hand.

SETAR, $h = 1$. You observe r_{T+1-d} , so you *know* which regime tomorrow is in:

$$\hat{r}_{T+1|T} = \phi_0^{(j)} + \sum_i \phi_i^{(j)} r_{T+1-i}, \quad j = \mathbf{1}\{r_{T+1-d} > c\} + 1.$$

No uncertainty about the regime. Just plug in.

The day after is not

SETAR, $h = 2$. The regime at $T + 2$ depends on r_{T+2-d} , which is still *random*. So

$$\mathbb{E}[r_{T+2} \mid \mathcal{F}_T] = \int f(r_{T+1}) \mathrm{d}F(r_{T+1} \mid \mathcal{F}_T) \neq f(\hat{r}_{T+1|T}).$$

The trap: Jensen's inequality

$\mathbb{E}[f(X)] \neq f(\mathbb{E}[X])$ whenever f bends.

Pushing the point forecast through the SETAR equation is **wrong**.

Plainly: *the average of a bent function is not the function of the average*.

It also throws away the interesting object: the predictive *distribution*, a mixture.

So simulate: let the model roll its own dice

For $m = 1, \dots, M$ (say $M = 10,000$)

- ① Draw shocks $\varepsilon_{T+1}^{(m)}, \dots, \varepsilon_{T+h}^{(m)}$ — either $N(0, 1)$, or resampled from the standardised residuals.
- ② Push them through the *fitted* SETAR, **re-checking which regime you are in at every step**.
- ③ Store the whole path.

Then $\hat{r}_{T+h|T} = \frac{1}{M} \sum_m r_{T+h}^{(m)}$, and the empirical quantiles of $\{r_{T+h}^{(m)}\}$ give the interval and the VaR.

Resampling residuals (rather than drawing normals) carries the fat tails of the S&P 500 into your intervals. Resampling the parameter estimates too carries estimation uncertainty. Each is one extra line of code.

Markov switching: forecasting the mood

The idea

Forecasting the mood is just repeated multiplication by P .

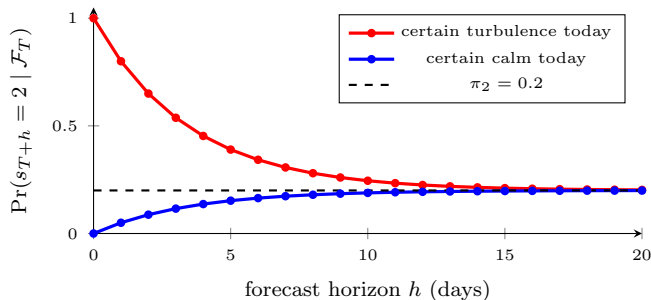
$$\xi_{T+h|T} = (P')^h \xi_{T|T} \xrightarrow{h \rightarrow \infty} \pi.$$

Then the return forecast is a probability-weighted blend of the two worlds:

$$\hat{r}_{T+h|T} = \sum_{j=1}^2 \xi_{T+h|T,j} \mu_j$$

Far enough ahead, $\xi_{T+h|T} \rightarrow \pi$ and you have forgotten today entirely.

How fast does the model forget today?



With $p_{11} = 0.95$, $p_{22} = 0.80$, both starting beliefs converge to $\pi = (0.8, 0.2)$ at geometric rate $|\lambda_2| = p_{11} + p_{22} - 1 = 0.75$.

*Today's regime is informative for about ten days and worthless after thirty.
A one-year forecast from a Markov switching model is π with extra steps.*

Two sources of risk, not one

The predictive density is a two-component normal mixture, and its variance splits in two:

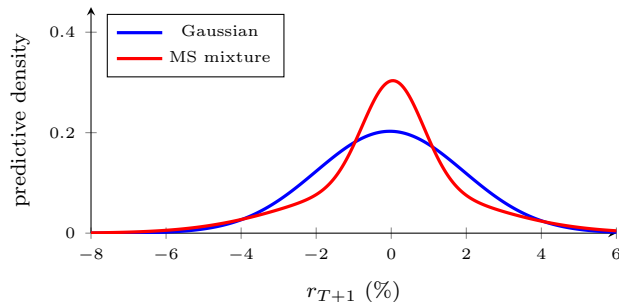
$$\text{Var}(r_{T+h} \mid \mathcal{F}_T) = \underbrace{\sum_j \xi_j \sigma_j^2}_{\text{uncertainty inside a world}} + \underbrace{\sum_j \xi_j (\mu_j - \hat{r}_{T+h|T})^2}_{\text{uncertainty about which world}} .$$

Why the second term matters

An MS model can produce a wide, skewed interval even when both regimes are perfectly homoskedastic.

Not knowing where you are is itself a source of risk.

Same mean, same variance, different risk



Both have mean -0.036 and variance 3.871 (from $\xi_{T+1|T} = (0.425, 0.575)$, $\mu = (0.05, -0.10)$, $\sigma = (0.8, 2.5)$).

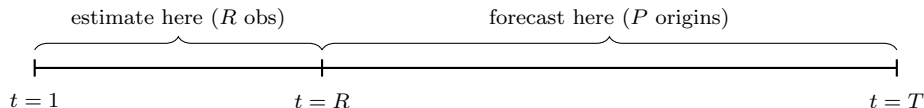
Identical MSFE. Identical Mincer–Zarnowitz regression. But the 1% quantile is -5.38 for the mixture and -4.61 for the Gaussian: a 17% difference in the capital you must hold.

Point-forecast evaluation is structurally blind to this.

Design the horse race before you run it

The idea

Almost every embarrassing forecasting result comes from letting the model peek at the future.



Recursive: the window grows. **Rolling:** fixed width, slides. **Fixed:** estimate once, never again.

Two rules that are easy to break

Rule 1: re-estimate *everything* at each origin

That means $\hat{c}$, the lag order, all of it.

Choosing p once on the full sample is look-ahead bias — and it is invisible in the output.

Rule 2: nonlinear models need long windows

Each regime must actually appear in the estimation window.

With a short window the crash regime never shows up, and your SETAR quietly becomes an AR.

Diebold–Mariano: is one model reliably better?

The idea

Take the two models' losses, subtract, and ask whether the difference is systematically non-zero. It is a t -test on a difference. Nothing more.

With $d_t = L(e_{1t}) - L(e_{2t})$, for example $L(e) = e^2$:

$$\text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{LRV}}(d_t)/P}} \xrightarrow{d} N(0, 1), \quad \widehat{\text{LRV}} = \hat{\gamma}_0 + 2 \sum_{k=1}^{h-1} \hat{\gamma}_k.$$

For $h > 1$ the differences are autocorrelated, hence the Newey–West term.

Harvey–Leybourne–Newbold rescale by $\sqrt{\{P + 1 - 2h + h(h - 1)/P\}/P}$ and use t_{P-1} .

When DM is illegal

The trap: nesting

An $AR(p)$ is a SETAR with $\phi^{(1)} = \phi^{(2)}$. It is *inside* the SETAR.

- If the AR is true, the two forecasts converge on each other.
- So $\bar{d} \rightarrow 0$ *and* its variance $\rightarrow 0$: you are dividing nothing by nothing.
- DM becomes degenerate and almost never rejects. “Fail to reject” means nothing.

Use Clark–West instead

It credits the big model for the estimation noise it was forced to carry:

$$\tilde{d}_t = (r_t - \hat{r}_{1t})^2 - [(r_t - \hat{r}_{2t})^2 - (\hat{r}_{1t} - \hat{r}_{2t})^2].$$

MSFE is the wrong scoreboard

The idea

You did not build a regime-switching model to shave 0.1% off the squared error of the mean. You built it to get the tails and the turning points right.

$$\begin{aligned} & \textit{Expect } \Delta MSFE \approx 0 \\ & \textit{and } \Delta \log\text{-score} > 0. \end{aligned}$$

That combination is a success, not a failure.

Scoring the point forecast

- **MSFE, MAE, and out-of-sample R^2 :**

$$R_{OS}^2 = 1 - \frac{\sum (r_t - \hat{r}_t)^2}{\sum (r_t - \bar{r}_t)^2}.$$

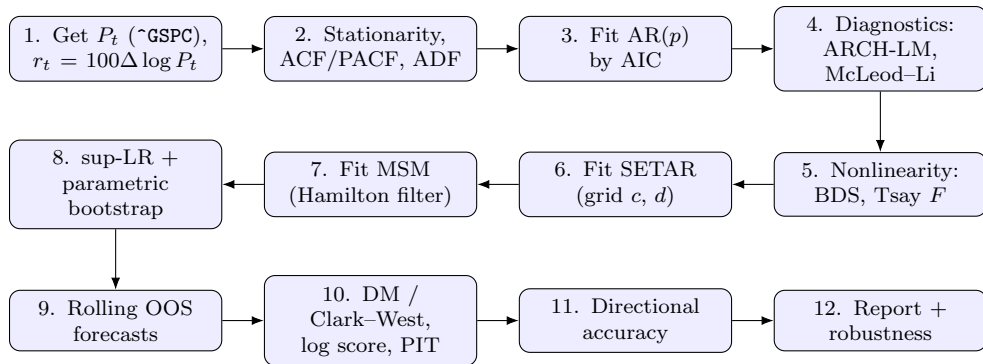
- **Directional accuracy.** Did you get the sign right? Tested formally by Pesaran–Timmermann. For returns this often matters more than MSFE.
- **Mincer–Zarnowitz.** Regress $r_t = \alpha + \beta \hat{r}_t + u_t$ and test $(\alpha, \beta) = (0, 1)$. Detects bias and systematic over- or under-reaction.

Scoring the density forecast

- **PIT.** $u_t = F_t(r_t)$ must be i.i.d. $U(0, 1)$.
A U-shaped histogram means your intervals are too narrow; a hump means too wide.
- **Log predictive score.** $\frac{1}{P} \sum \log f_t(r_t)$. Rewards getting the *shape* right, not just the centre.
- **CRPS** and **pinball loss**, when the quantity of interest is a quantile (VaR).
- **Model Confidence Set** (Hansen–Lunde–Nason) when comparing many models at once.

Nonlinear models earn their keep in the density and in the signs.

The whole pipeline



Implemented, in this order, in `sp500_nonlinear.R` and `sp500_nonlinear.Rmd`.

What you will find: the mean and the variance

- **The mean is nearly unpredictable.**

The AR(1) coefficient on daily data is small, often negative, barely significant.

- **ARCH-LM and McLeod–Li explode.**

Overwhelming rejections. That is nonlinearity in the *variance*.

- **BDS rejects on AR residuals, but barely on GARCH-standardised residuals.**

Translation: most of the “nonlinearity” was volatility all along.

What you will find: the models

- **Tsay F and sup-LR may reject** — until you regenerate the bootstrap null under AR–GARCH instead of AR with i.i.d. errors. Then the p -value often collapses.
- **Markov switching works robustly.**
It finds a long calm regime ($\hat{p}_{11} \approx 0.98\text{--}0.99$) and a short violent one. The dating lines up with 2008, 2020 and 2022 without being told to.
- **Forecasts:** MSFE gains over the AR are negligible. Density and VaR gains are real and large.

Discipline

Always report the bootstrap p -value *together with the null you simulated*. A threshold test is only as credible as its null.

Take away: the two models

- ① **TAR.** The regime is visible. Two regressions and a grid search. $\hat{c}$ is estimated unusually precisely — faster than everything else in the model.
- ② **Markov switching.** The regime is hidden. Predict, then Bayes-update, once per day. Out comes the likelihood *and* a crisis chronology.
- ③ **Separate mean nonlinearity from variance nonlinearity**, or you will fit a threshold to GARCH.

Take away: the inference

- ④ **You searched for c , so the χ^2 table is wrong.** Bootstrap — and re-run the *whole* search inside every replication.
- ⑤ **Multi-step forecasts need simulation (TAR) or matrix powers and a blend (MSM).** Never iterate a point forecast through a bent function.
- ⑥ **Use DM only for non-nested models;** Clark–West when nested. And grade the density, not just the mean.

Next: the problem set, then the R session.

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